STAT 2793 Homework 7 Solutions

1a) $x = 733$, $n = 1218$ $\Rightarrow$ $\hat{p} = 0.6018$

Check

$n\hat{p} = 733$, $n(1-\hat{p}) = 485$ $\Rightarrow$ Both > 5 ✓

95% CI: $\alpha = 0.05$

$\alpha/2 = 0.025$

$z_{0.025} = 1.96$

$$95\% \text{ CI}: \quad 0.6018 \pm 1.96\sqrt{\frac{0.6018(0.3982)}{1218}}$$

$$= (0.5743, \ 0.6293)$$

WE ARE 95% CONFIDENT THE TRUE PROPORTION OF STUDENTS WHO EXPECTED DIFFICULTY IN FINDING A JOB IS BETWEEN 57.43% AND 62.93%

b) $$ME = 1.96\sqrt{\frac{0.6018(0.3982)}{1218}}$$

$$= 0.0275$$

c) USE $$n = (0.25)\left(\frac{z_{\alpha/2}}{E}\right)^2$$

$$n = 0.25\left(\frac{1.96}{0.02}\right)^2$$

$$n = \underline{\underline{2401}}$$